

SenseGloves

Haptic gloves are one of the cornerstone research areas in the XR industry. Currently the only options for haptic gloves that exist cater to enterprises and businesses with the price tag ballooning into tens of thousands of dollars. SenseGloves are haptic gloves that bring down the cost relatively down, to a price tag of \$5000. SenseGloves are primarily focused on force-feedback rather than sensations. They employ a mix of cords that pull back on the fingers they move, imitating resistance, which can provide upto 20N of force. Sense gloves had a range of demos available at CES, from assembling a robot in VR to crushing a can. Nova appears to be a promising option for high-end business and institutional virtual reality at a far cheaper cost than existing options. Consumers, on the other hand, are reminded of how far this technology is from being inexpensive in mass market gadgets.



Vrgeiners' XTAL 3

Vrgeiners is currently at the forefront in the VR headset market in terms of technology with its newest mixed reality headset, XTAL 3. XTAL headsets have always been pushing boundaries when it comes to VR experience in terms of FOV, resolution and overall experience. XTAL 3 has a 180° horizontal and 90° vertical field of vision, which is presently the broadest visual field on the market for professional



XTAL 3

simulations. It provides an exceptional visual experience when used in conjunction with two 4K resolution screens and a pair of 4K mixed reality cameras. Most VR headsets with a FOV wider than 120 degrees suffer from some form of image distortion. However, advanced warping algorithms and lenses integrated in the XTAL 3 produce high-fidelity, undistorted pictures. Additional features include a patented eye-tracking technology with one-dot calibration for adjusting interpupillary distance with foveated rendering. The only catch is, the headset comes with a whopping \$11,500 price tag, and is only meant for industrial uses such as flight simulations.

ContactCI Gloves

ContactCI Gloves are a direct competitor to SenseGloves, albeit, with a slightly different use case. SenseGloves excel when it comes to the amount of force in its feedback, while ContactCI gloves focus on super-accurate finger-tip feedback with only a few newtons of force. It allows for very light sensations like flicking a switch, or pressing buttons. However, they're less of a pain to wear as compared to the SenseGloves. Again, these are also very enterprise focused, with consumer level products quite far away.



bHaptics TactGloves

bHaptics TactGloves is a consumer level product. It does not have any kind of force-feedback but rather just per finger precision motor that adds the missing feeling of touch lacking from other hand-tracking solutions like Quests' hand tracking. While TactGloves don't offer any other high level features like force-feedback, they can certainly add a dimension of immersion in the existing VR games. With its \$300 price tag, it's a very appealing product for early adopters of VR, especially considering that they're going to ship this year.

